Spatially Varying Prior Learning for Blind Hyperspectral Image Fusion

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***Abstract*— In recent years, researchers have become more interested in hyperspectral image fusion (HIF) as a potential alternative to expensive high-resolution hyperspectral imaging systems, which aims to recover a high-resolution hyperspectral image (HR-HSI) from two images obtained from low-resolution hyperspectral (LR-HSI) and high-spatial-resolution multispectral (HR-MSI). It is generally assumed that degeneration in both the spatial and spectral domains is known in traditional model-based methods or that there existed paired HR-LR training data in deep learning-based methods. However, such an assumption is often invalid in practice. Furthermore, most existing works, either introducing hand-crafted priors or treating HIF as a black-box problem, cannot take full advantage of the physical model. To address those issues, we propose a deep blind HIF method by unfolding model-based maximum a posterior (MAP) estimation into a network implementation in this paper. Our method works with a Laplace distribution (LD) prior that does not need paired training data. Moreover, we have developed an observation module to directly learn degeneration in the spatial domain from LR-HSI data, addressing the challenge of spatially-varying degradation. We also propose to learn the uncertainty (mean and variance) of LD models using a novel Swin-Transformer-based denoiser and to estimate the variance of degraded images from residual errors (rather than treating them as global scalars). All parameters of the MAP estimation algorithm and the observation module can be jointly optimized through end-to-end training. Extensive experiments on both synthetic and real datasets show that the proposed method outperforms existing competing methods in terms of both objective evaluation indexes and visual qualities.**

***Index Terms*— Hyperspectral image fusion (HIF), deep unfolding network, maximum a posteriori (MAP) estimation, Laplace distribution (LD) prior.**

I. Introduction

DUE to the rich spectral information, hyperspectral images (HSIs) have a better ability to identify the physical properties of target materials than standard multispectral images (MSIs), HSIs have achieved better results in various computer vision tasks, e.g., object detection [1], target recognition [2], target tracking [3], [4], [5], etc. However, hyperspectral imaging systems require a long exposure time to acquire many bands in the narrow spectral window at the same time, limiting their spatial resolution performance. Compared to RGB images, it is often more difficult to trade off spatial resolution and hyperspectral resolution due to physical limitations of the hardware and budget constraints.

Consequently, alternative computational solutions to working with low-resolution (LR) images and MSI images in hyperspectral imaging have attracted increasing attention, such as super-resolution [6], [7], fusion [8] and pan-sharpening [9]. Among them, the hyperspectral image fusion (HIF) method [10], [11], [12], [13], which fused a pair of LR-HSI and HR-MSI to obtain an HR-HSI, which has been widely used in hyperspectral imaging and has been shown to be highly effective. HIF aims to recover HR-HSI $\mathbf{Z} \in \mathbb{R}^{B\times N}$ from the corresponding LR-HSI $\mathbf{X} \in \mathbb{R}^{B\times n}$ and HR-MSI $\mathbf{Y} \in \mathbb{R}^{b\times N}$, where $N = W * H$, $n = w * h$ and $w \ll W$, $h \ll H$, and $b \ll B$ denote the widths, heights, and number of spectral bands of different images, respectively. In the traditional model-based method, the relationship between the observed low-resolution image pair and the original image is expressed as follows.

$$\mathbf{X} = \mathbf{ZHS}, \quad \mathbf{Y} = \mathbf{PZ}, \tag{1}$$

where $\mathbf{H}$ denotes the convolution operation (a.k.a. blurring kernel) followed by the spatial downsampling operation $\mathbf{S}$, and $\mathbf{P}$ stands for the spectral downsampling operation, usually defined as a linear model. Most existing HIF methods assume that degradations in the spatial and spectral domains are known [14], [15]. However, the degradation in practice is more complex than the predefined one and varies for each input LR-HSI, which will inevitably cause the mismatch between the assumed model and the real-world data, resulting in a performance decrease.

One promising remedy to the above problem is to investigate blind HIF, in which the degradation models are assumed to be unknown and image-independent. Existing blind HIF [11], [16], [17] methods can be roughly classified into two categories: model-based and learning-based methods. Model-based HIF methods used the observation model and a priori knowledge of HSIs to obtain satisfactory results. However, most model-based HIF methods assume that the degradation is

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known. To our knowledge, HySure [18] is the only conventional model-based method that can learn the observation model from input images. Despite the theoretical appeal, model-based methods cannot jointly optimize the parameters and rely on hand-crafted prior of HSIs, resulting in performance degradation.

Inspired by the success of deep learning, several blind HIF methods based on deep learning [11], [13], [16] have been proposed, the methods of [11], [13], and [16] reconstructed HR-HSI with a deep prior from paired LR-HSI and HR-MSI without known degradation models and corresponding HR-HSI. Nevertheless, they proposed to learn a degradation model to adapt to all data, rather than learning the degradation model from the corresponding input data, which did not meet the requirement that the degradation model varies from each input image. Their success achieved in blind HIF depends more on the great robustness of the proposed model, rather than the design that coincides with reality. Furthermore, due to failing to explore the physical degradation model deeply, ignored important information contained in physical degradation models needs to be compensated for by some pre-training operations (e.g., [11], [13]). Moreover, most existing deep learning-based methods often lack transparency (i.e., black-box design), resulting in difficulties in interpretation or optimization. To connect these two lines of research, there are some model-guided learning methods [12], [19], [20], [21], [22], which try to explore the physical model of hyperspectral images and unfold it into deep networks, leading to appreciable improvement.

The motivation behind our work is two-fold. On the one hand (adaptive kernel estimation), we propose to estimate the blurring kernels with a subnetwork for each input LR-HSI in blind HIF, while the spectral degradation, which depends on intrinsic parameters of the camera, will be learned through end-to-end optimization. On the other hand (prior learning), the MAP estimation of HR-HSIs with the learned Laplace Distribution (LD) prior, which implicitly introduces a sparse regularization [23], has been unfolded into a deep neural network, where both the priors of HR-HSIs and their uncertainty (i.e., the variance of the prior) were jointly learned by a deep network. To better exploit the spatio-spectral correlations of HSIs, we propose to borrow the Swin-Transformer [24] to estimate both the prior (i.e., the means) and the uncertainty (i.e., the variances) of the unknown HR-HSI image. Extensive experimental results have shown that the proposed method outperforms several existing state-of-the-art blind HIF methods. The key contributions of this paper are summarized as follows:

- The learned LD models are proposed to leverage the spatial-spectral correlations of the HSIs, while the sparse regularization is introduced into our method implicitly.
- Instead of learning a uniform prior for unknown HR-HSIs which is commonly used in previous works, the uncertainty (variances) of the priors which are spatially varying, is also learned by a novel application of Swin-Transformer modules adaptively.
- We propose to learn the input-dependent spatial blur kernels $\mathbf{H}$ from the input LR-HSIs through a specially designed CNN. For the first time, we attempt to address the issue that degradation kernels, which are camera-dependent, often vary from each input.

II. Related Works

In this section, we briefly review blind HIF methods, including model-based methods and deep learning-based methods, unsupervised HIF methods, and transformer-based image restoration.

A. Blind HIF

While most model-based HIF methods were nonblind, the HySure method [18] estimated the degradation models from the input images by solving a convex optimization problem. However, the hand-crafted prior leads to a cumbersome design and disables jointly optimizing parameters. Several learning-based methods [11], [12], [16], [17], [25] have also been proposed for blind HIF, achieving promising results in recent years. Due to the unknown degradation model, blind HIF was required to learn the degradation model during training. Wang et al. [25] proposed to estimate the degradation models for HIF iteratively. Methods of [11] were proposed to pre-determined and adaptively learn the two observation models for blind HIF. In [16], an MS / HS fusion model was implemented that merges the physical models of low-resolution images and the low-ranking prior knowledge of HR-HSI in the deep learning network, in which the degradation model was also included. More recently, MoG-DCN [12] developed the physical model into an iterative optimization algorithm, allowing the degradation model to be learned through end-to-end training.

However, all of these methods assumed that the spatial degradation models were constant for all input LR-HSIs, leading to the problem that a trained network contains a stationary degradation model, which was unrealistic for practical applications.

B. Unsupervised HIF

Unsupervised HIF aimed at training the HIF without any label (HR-HSI), which was based primarily on the physical model. Reference [20] used an unsupervised encoder-decoder architecture composed of two encoder-decoder networks, which is encouraged to follow a sparse Dirichlet distribution to realize unsupervised HIF. Similarly, Yao et al. [19] proposed a coupled unmixing network with a cross-attention mechanism for unsupervised HIF. The UAL proposed in [13] first implicitly learns a general image prior with deep networks and then adapts it to a specific HSI.

Although they all utilize the information of the physical model more or less, few of them took into account the deep information of the physical model. In addition, to solve the highly unconstrained problem, some of them [19] minimize multiple domain-specific loss functions, making it difficult to select optimal hyperparameters.

C. Transformer-Based Image Restoration

Methods based on transformer models, first proposed for NLP, have outperformed many convolution-based methods in many image tasks, e.g., image classification [26], [27], [28] and image reconstruction [29], [30], [31]. However, simply replacing convolution layers with transformer layers will cause a large computational burden. Inspired by the promising performance and the significant reduction in computing and memory consumption of the Swin-Transformer [24], a U-shaped architecture [32] based on the Transformer has been proposed for image reconstruction, which has fewer parameters and fewer computations than U-Net. Recently, Swin-Transformer was used for various image reconstruction tasks [30], achieving state-of-the-art results.

III. Spatially-Varying Prior For Hyperspectral Image Fusion

A. Problem Formulation With LD Model

To solve the highly ill-posed inverse problem of recovering HR-HSI from a pair of LR-HSI and HR-MSI, we propose to formulate the reconstruction of the HIF image as a MAP estimation problem. Given the observed images ($\mathbf{X}$ and $\mathbf{Y}$), the HR-HSI target image $\mathbf{Z}$ can be estimated along with $\mathbf{H}$ and $\mathbf{P}$ by

$$p(\mathbf{H}, \mathbf{P}, \mathbf{Z}|\mathbf{X}, \mathbf{Y}) \propto p(\mathbf{H}|\mathbf{X}, \mathbf{Y})p(\mathbf{P}|\mathbf{X}, \mathbf{Y})p(\mathbf{Z}|\mathbf{H}, \mathbf{P}, \mathbf{X}, \mathbf{Y}). \tag{2}$$

By maximizing the logarithmic version of the posterior of Eq. (2), the HR-HSI and degradation models can be obtained by minimizing the following objective function.

$$\begin{aligned}(\mathbf{H}^*, \mathbf{P}^*, \mathbf{Z}^*) = \arg\max_{\mathbf{H},\mathbf{P},\mathbf{Z}}[\log p(\mathbf{H}|\mathbf{X}, \mathbf{Y}) + \log p(\mathbf{P}|\mathbf{X}, \mathbf{Y})\\ + \log p(\mathbf{H}, \mathbf{P}, \mathbf{X}, \mathbf{Y}|\mathbf{Z}) + \log p(\mathbf{Z})],\end{aligned} \tag{3}$$

which can be iteratively solved by alternatively solving the following three subproblems.

$$\hat{\mathbf{H}} = \arg\max_{\mathbf{H}} \log p(\mathbf{H}|\mathbf{X}, \mathbf{Y}), \tag{4}$$

$$\hat{\mathbf{P}} = \arg\max_{\mathbf{P}} \log p(\mathbf{P}|\mathbf{X}, \mathbf{Y}), \tag{5}$$

$$\hat{\mathbf{Z}} = \arg\max_{\mathbf{Z}} \log p(\mathbf{H}, \mathbf{P}, \mathbf{X}, \mathbf{Y}|\mathbf{Z}) + \log p(\mathbf{Z}). \tag{6}$$

Among them, we propose to solve the $\mathbf{H}$-subproblem and $\mathbf{P}$-subproblem using a deep learning-based subnetwork. In particular, the $\mathbf{H}$-subproblem is addressing the issue, which we will elaborate later. For the $\mathbf{Z}$-subproblem, $p(\mathbf{H}, \mathbf{P}, \mathbf{X}, \mathbf{Y}|\mathbf{Z})$ in Eq. (6) which is the likelihood term that can be further modeled by the following Gaussian distribution [33]

$$p(\mathbf{H}, \mathbf{P}, \mathbf{X}, \mathbf{Y}|\mathbf{Z}) \propto \exp(-\frac{||\mathbf{X} - \mathbf{ZHS}||_F^2}{2\sigma_{lr}^2} - \frac{||\mathbf{Y} - \mathbf{PZ}||_F^2}{2\sigma_{rgb}^2}), \tag{7}$$

where σ_{lr} and σ_{rgb} denote the variance of $\mathbf{X}$ and $\mathbf{Y}$, respectively, which are generally implemented as hyperparameters [34] or learnable scalars [12], [21]. For non-blind HIF, degradation is given in spatial and spectral domains; therefore, it is reasonable to treat every pixel in degraded images equally, as described in the equation. (7). However, it is still doubtful whether the likelihood term is completely accurate in blind HIF, which has never been taken into account in previous deep learning-based works. To address this issue, we prefer to describe the likelihood term in both the spatial and spectral domains with variances of $\mathbf{X}$ and $\mathbf{Y}$ in the pixel domain, which can also be described as a non-i.i.d. non-zero mean Gaussian probability model. Changing $\frac{1}{2\sigma_{lr}^2}$ and $\frac{1}{2\sigma_{rgb}^2}$ in Eq. (7) with $\boldsymbol{\eta}_{lr}$ and $\boldsymbol{\eta}_{rgb}$ pixel by pixel, Eq. (7) can be written further as

$$\begin{aligned}p(\mathbf{H}, \mathbf{P}, \mathbf{X}, \mathbf{Y}|\mathbf{Z}) \propto \exp(-\boldsymbol{\eta}_{lr}||\mathbf{X} - \mathbf{ZHS}||_F^2\\ - \boldsymbol{\eta}_{rgb}||\mathbf{Y} - \mathbf{PZ}||_F^2).\end{aligned} \tag{8}$$

Note that for the prior term in Eq. (6), a *nonzero*-mean Laplace distribution can be used to model each pixel z_i of $\mathbf{Z}$, which was introduced in image denoising and has been shown to be effective in [35]. With the assumption that z_i and θ_i are independent, we can characterize $\mathbf{Z}$ using the following model.

$$p(\mathbf{Z}) = \prod_i p(z_i),\ p(z_i) = \int_0^\infty p(z_i|\theta_i)p(\theta_i)d\theta_i, \tag{9}$$

where $p(z_i|\theta_i)$ is a Laplace distribution parameterized by the mean u_i and standard variance θ_i - i.e.,

$$p(z_i|\theta_i) = \frac{1}{2\theta_i}\exp(-\frac{|z_i - u_i|}{\theta_i}). \tag{10}$$

It should be mentioned that the parameters u_i and θ_i introduced by Eq. (10) characterize the probability model of each pixel z_i, setting the stage for this work.

B. Optimization for the MAP Estimator

The prior of the scale parameter $p(\theta_i)$ can be modeled with an energy function, i.e., $p(\theta_i) \propto exp(-J(\theta_i))$. Then we can convert Eq. (6) into the following dual-variable MAP estimation problem.

$$(\hat{\mathbf{Z}}, \hat{\boldsymbol{\theta}}) = \arg\max_{\mathbf{Z},\boldsymbol{\theta}} \log p(\mathbf{H}, \mathbf{P}, \mathbf{X}, \mathbf{Y}|\mathbf{Z}) + \log p(\mathbf{Z}|\boldsymbol{\theta}) - J(\boldsymbol{\theta}). \tag{11}$$

By substituting the data likelihood term of Eq. (7) and the prior term of Eq. (10) into the MAP estimation, we can obtain the following result.

$$\begin{aligned}(\hat{\mathbf{Z}}, \hat{\boldsymbol{\theta}}) = \arg\min_{\mathbf{Z},\boldsymbol{\theta}} \boldsymbol{\eta}_{lr}||\mathbf{X} - \mathbf{ZHS}||_F^2 + \boldsymbol{\eta}_{rgb}||\mathbf{Y} - \mathbf{PZ}||_F^2\\ + \sum_{i=1}^N(\frac{1}{2\theta_i}|z_i - u_i| + \log\theta_i) + J(\boldsymbol{\theta}),\end{aligned} \tag{12}$$

where $\boldsymbol{\eta}_{lr} \in \mathbb{R}^{B\times n}$ and $\boldsymbol{\eta}_{rgb} \in \mathbb{R}^{b\times N}$ carry the variance information of $\mathbf{X}$ and $\mathbf{Y}$, which is suggested to be learned from the deviation between given degraded images and estimated degraded images (that is, $\mathbf{X} - \mathbf{ZHS}$ and $\mathbf{Y} - \mathbf{PZ}$) in this work. To solve the optimization problem, we can alternately optimize $\mathbf{Z}$ and $\boldsymbol{\theta}$. For the $\mathbf{Z}$-subproblem, with fixed $\boldsymbol{\theta}$, we can solve $\mathbf{Z}$ by

$$\begin{aligned}\hat{\mathbf{Z}} = \arg\min_{\mathbf{Z}} \boldsymbol{\eta}_{lr}||\mathbf{X} - \mathbf{ZHS}||_F^2 + \boldsymbol{\eta}_{rgb}||\mathbf{Y} - \mathbf{PZ}||_F^2\\ + ||\mathbf{W} \odot |\mathbf{Z} - \mathbf{U}|||_1,\end{aligned} \tag{13}$$

where $\mathbf{W} \in \mathbb{R}^{B\times N}$ and $\mathbf{U} \in \mathbb{R}^{B\times N}$ are the matrix representations of $w_i = \frac{1}{2\theta_i}$ and u_i, respectively, and $\odot$ denotes the point-wise product operator. As shown in Eq. (13), modeling each pixel of $\mathbf{Z}$ using a Laplace distribution (LD) allows us to take advantage of the sparsity prior for a more realistic estimate than the previous work [36].

Note that the term $\mathbf{U}$ varies spatially in Eq. (13) served as a priori of the unknown $\mathbf{Z}$ can be estimated from the intermediate estimates of $\mathbf{Z}$. In general, the estimation of $\mathbf{U}$ can be interpreted as a denoising problem with respect to $\mathbf{Z}$. Specifically, the mean $\mathbf{U}$ can be calculated using a learning-based denoising module from $\mathbf{Z}$, which can be denoted by $\mathbf{U} = f_U(\mathbf{Z})$, where $f_U(\cdot)$ stands for learning-based denoising [37], [38].

Furthermore, instead of using a global regularization weight parameter for the prior regularization term with respect to $\mathbf{U}$ to evaluate the contribution that the learned prior can make to reconstruction, we propose imposing a locally adaptive regularization on $\mathbf{Z}$ with weights $\mathbf{W}$, which is computed with variances (uncertainty) of $\mathbf{Z}$. In some way, the incorporation of learned variances allows us to automatically adjust the contribution of the physical model and the prior learned on a pixel-by-pixel basis instead of a simple scalar in [39]. With the estimated $\mathbf{W}$ (it can also be regarded as prior), pixels z_i with larger variances θ_i will be imposed less strength of regularization and vice versa. Intuitively, the regularization term $||\mathbf{Z}-\mathbf{U}||_1$ will be minimized for pixels z_i whose variances θ_i tend toward 0, i.e., the solution will be u_i, on the contrary, the solution tends to minimize the data fidelity term when the variances θ_i of pixels z_i are large enough. As such, the proposed method will pay more attention to the edges and textures and a better estimation of $\mathbf{Z}$ can be achieved. The $\mathbf{Z}$-subproblem as described in Eq. (13) which is convex can be solved using an iterative shrinkage algorithm similar to [23]. In the $(t + 1)$-th iteration, the proposed shrinkage operator for $\mathbf{Z}$ is

$$\mathbf{Z}^{(t+1)} = S_\tau(\mathbf{V}^{(t)} - \mathbf{U}^{(t)}) + \mathbf{U}^{(t)} \tag{14}$$

where $S_\tau(\cdot)$ is the classic soft-thresholding operator given by

$$S_\tau(t) = \begin{cases} t+\tau & t < -\tau, \\ 0 & -\tau < t < \tau, \\ t-\tau & t > \tau, \end{cases} \tag{15}$$

$\tau = \mathbf{W}/c$ and c is an auxiliary parameter that guarantees the convexity of the surrogate function above; for simplicity, $1/c$ will be written as δ. As an intermediate auxiliary variable, $\mathbf{V}^{(t)}$ can be rewritten as

$$\mathbf{V}^{(t)} = \mathbf{Z}^{(t)} + \frac{1}{c}\{\mathbf{P}^\mathsf{T}\boldsymbol{\eta}_{rgb}(\mathbf{Y}-\mathbf{P}\mathbf{Z}^{(t)}) + \boldsymbol{\eta}_{lr}(\mathbf{X} - \mathbf{Z}^{(t)}\mathbf{H}\mathbf{S})(\mathbf{H}\mathbf{S})^\mathsf{T}\}, \tag{16}$$

The derivation of the above shrinkage operator follows the standard surrogate algorithm in [40]. Similarly, the $\boldsymbol{\theta}$-subproblem can turn into the problem of estimating $\mathbf{W}$. With a fixed $\mathbf{Z}$, $\mathbf{W}$ can be estimated by solving the following optimization problem.

$$\hat{\mathbf{W}} = \arg\min_{\mathbf{W}} \sum_{i=1}^{N}[w_i|z_i - u_i| - \log w_i] + J(\boldsymbol{\theta}). \tag{17}$$

Instead of introducing a hand-crafted prior and then solving the Eq. (16) in closed form or gradient descent algorithm, which usually causes the problem that all the parameters of the network cannot be jointly optimized, we propose a DCNN-based method (i.e. $f_{\mathbf{W}}(\cdot)$) to learn $\mathbf{W}^{(\mathbf{t})}$ from $\mathbf{Z}^{(\mathbf{t})}$ as will be described in detail in the next section.

Algorithm 1 Proposed Iterative HIF Algorithm

Input: The LR-HSI, $\mathbf{X}$ and the HR-MSI, $\mathbf{Y}$
Initialization:
(1) Set parameters λ, ϕ and t to 0
(2) Initialize $\mathbf{Z}^{(0)}$ to the bicubic interpolation of the corresponding LR-HSI $\mathbf{X}$
While does not converge **do**
(1) Estimate $\mathbf{H}$ from the given LR-HSI and $\mathbf{P}$
(2) Compute $\mathbf{U}^{(t)} = f_U(\mathbf{Z}^{(t)})$, $\mathbf{W}^{(t)} = f_W(\mathbf{Z}^{(t)})$
(3) Update $\mathbf{V}^{(t+1)}$ via Eq. (16)
(3) Update $\mathbf{Z}^{(t+1)}$ via Eq. (14)
(4) $t = t + 1$
Output: The estimated HR-HSI, $\mathbf{Z}$

The proposed iterative HIF algorithm is summarized in Algorithm 1. Furthermore, inspired by recent work, we propose to unfold Algorithm 1 into a DCNN-based implementation by bridging the $\mathbf{Z}$ and $\mathbf{W}$-subproblems via a unified framework -i.e.,

$$\mathbf{Z}^{(t+1)} = S_{\delta f_{\mathbf{W}}(\mathbf{Z}^{(t)})}(\mathbf{V}^{(t)} - f_{\mathbf{U}}(\mathbf{Z}^{(t)})) + f_{\mathbf{U}}(\mathbf{Z}^{(t)}), \tag{18}$$

where $f_W(\cdot)$ denotes a convolution network for estimating $\mathbf{W}^{(t)}$ from $\mathbf{Z}^{(t)}$, and $\mathbf{U}^{(t)}$ is also estimated by another convolution network -i.e., $\mathbf{U}^{(t)} = f_U(\mathbf{Z}^{(t)})$. Such a model-guided network design enjoys not only flexibility (all hyperparameters including prior-learning model related and physical-imaging model related can be tuned by end-to-end optimization) but also transparency (interpreted as a numerical solution to the well-defined optimization problem).

IV. Deep Network Implementation

A. The Overall Network Architecture

The overall network architecture of the proposed method is shown in Fig. 1 (a), which contains T stages corresponding to T iterations of Algorithm 1. As shown in Fig. 1 (a), the input LR-HSIs are first fed into the observation module to learn the corresponding blur kernels in the spatial domain, which play an important role in the following reconstruction module. At the same time, a bicubic upsample operation is applied in LR-HSIs $\mathbf{X}$ for the initial estimate $\mathbf{Z}^{(0)}$. Then, the estimated current $\mathbf{Z}^{(t)}$ and the input LR-HSIs $\mathbf{X}$ and HR-MSIs $\mathbf{Y}$ are fed into the reconstruction module to minimize the reconstruction error.

$$\mathbf{V}^{(t)} = \mathbf{Z}^{(t)} + \delta\{\mathbf{P}^\mathsf{T}\boldsymbol{\eta}_{rgb}(\mathbf{Y} - \mathbf{P}\mathbf{Z}^{(t)}) + \boldsymbol{\eta}_{lr}(\mathbf{X} - \mathbf{Z}^{(t)}\mathbf{H}\mathbf{S})(\mathbf{H}\mathbf{S})^\mathsf{T}\} \tag{19}$$

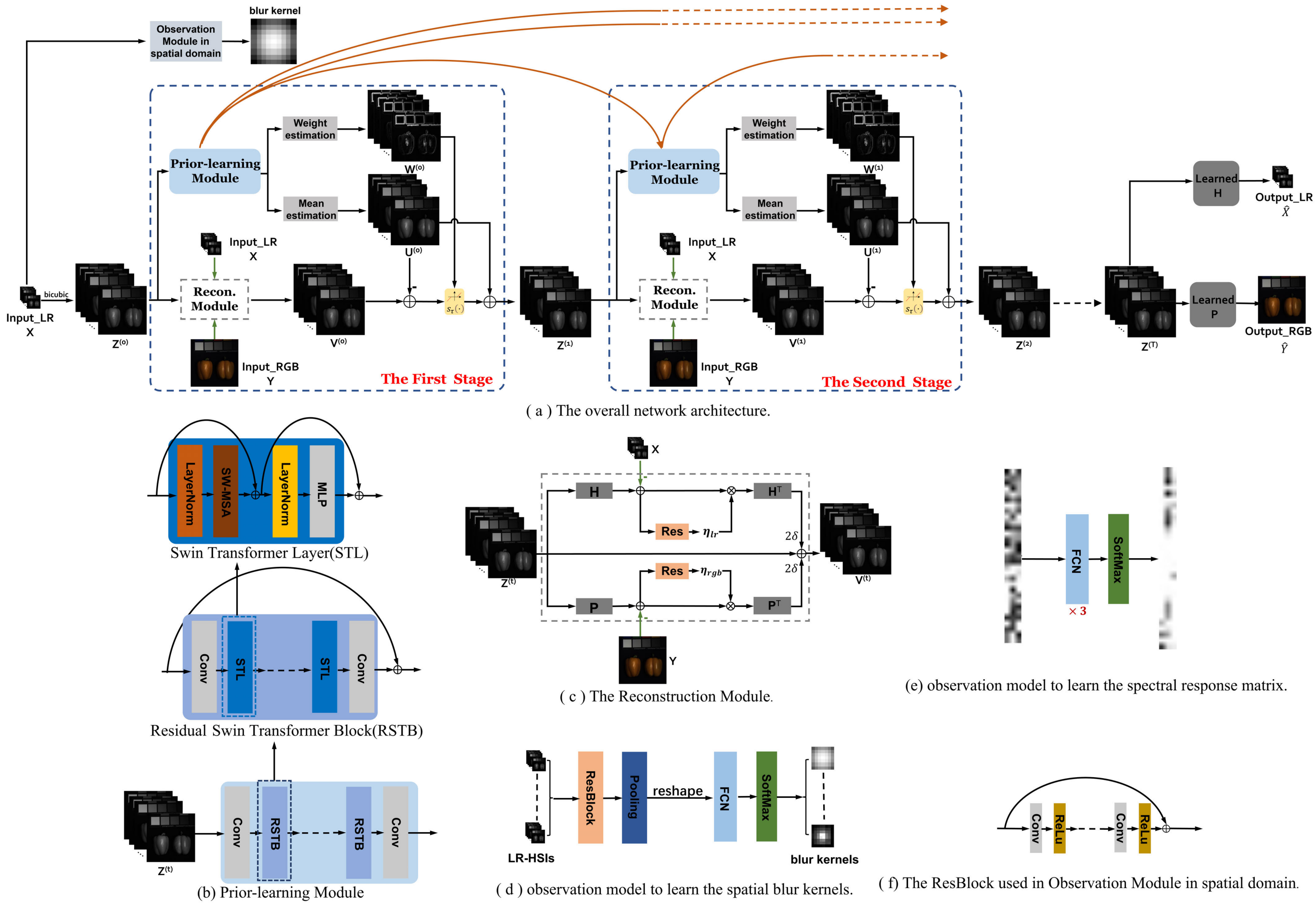


Fig. 1. Architecture of the proposed network for HIF. (a) The overall network architecture, (b) the Prior-learning Module, (c) the Reconstruction Module, (d) observation model to learn the spatial blur kernels, (e) the observation model to learn the spectral response matrix, and (f) the ResBlock used in Observation Module in the spatial domain.

Meanwhile, the mean prior $\mathbf{U}^{(t)}$ and the adaptive weight prior $\mathbf{W}^{(t)}$ learned from $\mathbf{Z}^{(t)}$ through a TPN(Transformer-Prior Net), as described in Fig.1 (a), along with the middle estimated terms $V^{(t)}$ are fed into a soft-thresholding operation module as shown in Eq. (14). After T iterations, the learned degradation operations are applied to the latent HR-HSIs so that we can obtain the degraded images (estimated LR-HSIs and HR-MSIs) needed for the unsupervised task.

Note that a similar unfolding strategy has been proven to be successful in image restoration [37], [38]. However, what makes our work different from previous model-based works is the design of adaptive weight with regard to the learned priors in every pixel, aiming at addressing the issue that the contribution of the learned deep priors is estimated coarsely caused by the design of a simple global trade-off parameter in previous works.

B. Observation Module

The objective of the observation network is to estimate the blur kernels $\mathbf{H}$ for all input LR-HSIs, and it is assumed that the spectral degradation matrix $\mathbf{P}$ is constant for all input HR-MSIs, as shown in Figs. 1 (d) and (e). It is generally assumed that both $\mathbf{H}$ and $\mathbf{P}$ are sparse, non-negative, and would sum up to 1 (normalization constraint). Due to the correlation between $\mathbf{H}$ and the LR-HSI, we propose learning $\mathbf{H}$ from the corresponding LR-HSI through a CNN, as shown in Fig. 1(d). The input LR-HSI first goes through three res-blocks followed by a pooling layer to encode the image, then several fully connected layers are applied to decode the feature, which is followed by a SoftMax nonlinearity layer to guarantee the non-negative and normalization constraints. Unlike image-dependent $\mathbf{H}$, $\mathbf{P}$ depends on camera parameters, which tend to be constant in the same data set. Therefore, several fully connected layers and one softMax nonlinearity layer are applied with a random initialized matrix to simulate $\mathbf{P}$. In our current implementation, the size of the spatial blurring kernel is set to 14×14, and the size of the spectral response matrix is set to 31×3 in the CAVE and Harvard datasets. In previous work, several loss functions were designed to satisfy the constraints mentioned above. Our experimental results show that the introduction of softMax nonlinearity is sufficient to guarantee these constraints. Furthermore, we apply a deconvolution in every latent LR-HSI with the parameters of the learned $\mathbf{H}$ to implement $\mathbf{H}^\top$ and multiply the transposition of the matrix $\mathbf{P}$ with the input HR-MSI to realize $\mathbf{P}^\top$.

C. Reconstruction Module

The reconstruction module aims to compute the intermediate estimate $\mathbf{V}^{(t)}$ as described in Eq. (14). As shown in Fig. 1(c), the architecture of the reconstruction module mainly involves degradation operators in both spatial and spectral domains (i.e. $\mathbf{H}$ and $\mathbf{P}$), their transposed versions (i.e., $\mathbf{H}^{\mathsf{T}}$ and $\mathbf{P}^{\mathsf{T}}$) and the spatially varying weight applied in residual error between the estimated degraded image and the ground truth in both spatial and spectral domains (i.e., $\boldsymbol{\eta}_{lr}$ and $\boldsymbol{\eta}_{rgb}$). The symmetric architecture states the dual role that the degraded images play in the reconstruction module, while the weight design $\boldsymbol{\eta}_{lr}$ and $\boldsymbol{\eta}_{rgb}$ based on the variance of the spatial and spectral degraded images try to assess the credibility of the estimated degraded image.

More specifically, we propose feeding the residual error between the estimated degraded image and ground truth into a Resblock containing three convolutional layers to obtain the weights $\boldsymbol{\eta}_{lr}$ and $\boldsymbol{\eta}_{rgb}$, and then a point-wise product operator will be applied to the learned weights and the residual errors from two parallel channels, respectively, corresponding to Eq. (14). Details about how to generate the blur kernel and the spectral degradation matrix and how to implement the operators $\mathbf{H}$ and $\mathbf{P}$ with degradation information are given in Sec. IV-B.

D. Transformer-Based Prior Network

In previous work such as DBIN [25] and MHF-net [41], ResNet was used to learn a proximal operator and achieved promising performance. Recently, MoG-DCN [12] and DHIF [21] proposed an Unet [42]-based denoiser to learn the prior, which achieves state-of-the-art (SOTA) performance on nonblind HIF. Inspired by the latest advances in computer vision [24], we propose to design a transformer-based prior learning network. The Transformer-based Prior Network(TPN) is aimed to learn the adaptive mean prior $\mathbf{U}^{(t)}$ and weight prior $\mathbf{W}^{(t)}$ learned from the estimated current $\mathbf{Z}^{(t)}$, i.e.

$$\mathbf{U}^{(t)} = \mathrm{CNN_u}(\mathrm{TPN}(\mathbf{X}^{(t)})),$$
$$\mathbf{W}^{(t)} = \mathrm{CNN_w}(\mathrm{TPN}(\mathbf{X}^{(t)})), \tag{20}$$

where $\mathrm{TPN}(\cdot)$, $\mathrm{CNN_u}(\cdot)$ and $\mathrm{CNN_w}(\cdot)$ denote transformer prior network, the mean estimation module and the weight estimation module composed of CNN network.

The transformer prior network (TPN) used in the proposed method is shown in Figs. 2(a)-(b), whose main components include the Residual Swin Transformer Block (RSTB) and convolutional layers. RSTB is a residual block consisting of the Swin Transformer layer (STL) proposed in [24].

As shown in Figure 3 (b), the output of the Swin Transformer layer can be calculated as

$$\mathbf{X}_M^{(t)} = \mathrm{MSA}(\mathrm{LN}(\mathbf{X}^{(t)})) + \mathbf{X}^{(t)},$$
$$\mathbf{X}_F^{(t)} = \mathrm{FFN}(\mathrm{LN}(\mathbf{X}_M^{(t)})) + \mathbf{X}_M^{(t)}, \tag{21}$$

where $\mathbf{X}^{(t)}, \mathbf{X}_M^{(t)}, \mathbf{X}_F^{(t)} \in \mathbb{R}^{m^2\times c}$ denotes the local windows, $\mathrm{MSA}(\cdot)$ and $\mathrm{LN}(\cdot)$ denote the multi-head self-attention mechanism and the LayerNorm layer, and $\mathrm{FFN}(\cdot)$ denotes a multi-layer perceptron (MLP) consisting of two fully-connected layers and a GELU non-linearity layer.

The main difference between the Swin transform layer (STL) and the original transformer layer lies in the local attention and the shifted window mechanism, which allow long-range modeling capabilities with fewer computations. In addition, dense inter-stage connections are introduced to bridge the first feature maps of the prior network from the previous stage to the current stage.

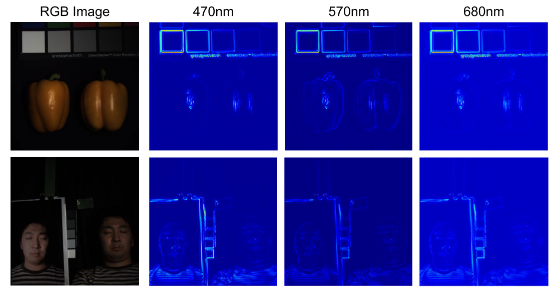


Fig. 2. Visualization of the regularization parameters $\mathbf{W}$ in the third stage. Left: RGB images; right: the corresponding $\mathbf{W}$ images in 3 spectral bands.

Finally, the result of the prior network is fed into the weight estimation and the mean estimation to generate intermediate estimated priors $\mathbf{W}^{(\mathbf{t})}$ and $\mathbf{U}^{(\mathbf{t})}$, respectively. Some visualization results of weight maps $\mathbf{W}$ in Fig. 2. From Fig. 2, we can see that the learned $\mathbf{W}$ spatially varies and is faithful to the edges and textures of the images, which shows that the network we proposed can capture the information around the edges, leading to better reconstruction results.

E. Network Training

One of the key challenges in our model is the requirement of many paired training data for supervised learning, while the image-varying blur kernels learned in our model play the role of a bridge between degraded images and latent reconstructed images. Therefore, we naturally introduce an unsupervised framework (without knowing $\mathbf{Z}$ in the training stage), leading to the following sum as the surrogate reconstruction loss.

$$\mathbf{L}_{rec} = ||\hat{\mathbf{Z}}\mathbf{HS} - \mathbf{X}||_1 + ||\mathbf{P}\hat{\mathbf{Z}} - \mathbf{Y}||_1 \tag{22}$$

where $\hat{\mathbf{Z}}\mathbf{HS}$ and $\mathbf{P}\hat{\mathbf{Z}}$ denote the reconstructed LR-HSI and HR-MSI images, respectively, and $\hat{\mathbf{Z}}$ can be written as $\hat{\mathbf{Z}} = \mathcal{F}(\mathbf{X}, \mathbf{Y}; \Theta, \delta)$, where Θ represents the learnable parameters of the modules, including the observation module, the prior network based on transformers, δ is the step parameter of learnability that differs from each stage. We have adopted L_1 instead of L_2 loss due to its improved robustness to large errors, as well as perceptually satisfying performance in low-level vision tasks. Furthermore, inspired by the consistency loss of CUCaNet [19], we introduce a local spatio-spectral consistency term to better characterize the observation model of Eq. (14) as follows.

$$\mathbf{L}_{con} = ||\mathbf{YHS} - \mathbf{PX}||_1 \tag{23}$$

Note that the consistency loss is faithful to the actual physical models, which contributes to learning more accurate degradation models. By integrating the two loss functions above, the final loss function can be expressed as

$$\mathbf{L} = \mathbf{L}_{rec} + \alpha\mathbf{L}_{con} \tag{24}$$

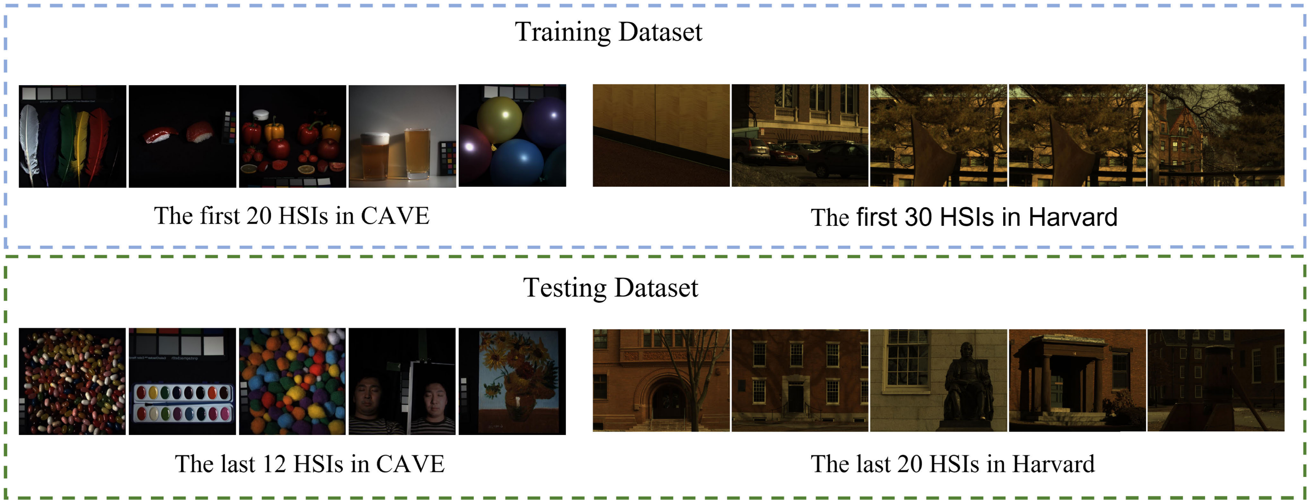


Fig. 3. The overview of the used datasets in simulation data experiments, we show the corresponding RGB images of HSIs.

TABLE I

AVERAGE PSNR, SAM, ERGAS, AND SSIM RESULTS OF THE TEST METHODS ON THE CAVE DATASET AND THE HARVARD DATASET

	HySure [18]	NSSR [6]	DBSR [11]	CuCANet [19]	DBIN [25]	UAL [13]	Ours(Unet)	Ours(Transformer)
s=8(CAVE)								
PSNR(dB)	36.31	41.20	40.19	39.97	38.69	40.81	43.56	**44.22**
SAM	9.63	4.01	4.84	5.06	5.53	3.92	3.36	**3.28**
ERGAS	2.11	0.93	1.04	1.16	1.19	0.99	0.80	**0.78**
SSIM	0.903	0.978	0.974	0.966	0.964	0.985	0.994	**0.995**
s=16(CAVE)								
PSNR(dB)	33.18	38.62	38.71	38.99	38.02	38.55	39.60	**40.39**
SAM	11.85	5.46	5.48	5.62	5.93	5.69	5.39	**5.01**
ERGAS	1.46	0.70	0.81	0.83	0.87	0.64	0.61	**0.57**
SSIM	0.889	0.962	0.966	0.969	0.957	0.974	0.981	**0.984**
s=8(Harvard)								
PSNR(dB)	38.31	41.79	39.92	40.37	38.69	41.01	42.48	**42.93**
SAM	6.13	4.71	5.46	5.06	5.53	4.92	4.52	**4.31**
ERGAS	1.98	**1.26**	1.56	1.42	1.54	1.44	1.29	1.27
SSIM	0.945	0.978	0.969	0.972	0.964	0.985	0.989	**0.989**
s=16(Harvard)								
PSNR(dB)	36.77	**40.23**	39.04	38.38	37.92	39.25	39.60	40.08
SAM	6.54	**4.96**	5.65	5.65	5.88	5.48	5.44	5.13
ERGAS	1.27	0.67	0.83	0.86	0.91	0.69	0.73	**0.66**
SSIM	0.933	0.964	0.957	0.960	0.947	0.964	0.965	**0.970**

where α is the parameter that controls the trade-off between the two loss terms. With the loss function designed above, the network parameters Θ and the parameter δ can be jointly optimized by end-to-end training.

V. Experimental Results

A. Datasets and Training Setting

In this study, two widely used HSI datasets were used to evaluate the performance of the proposed method, i.e., the CAVE [43] and Harvard datasets [44]. The CAVE dataset contains 32 indoor HSIs that are collected by a variety of pixel cameras. Each image in the CAVE has 512×512 pixels in the spatial domain and contains 31 spectral bands that span 400 to 700 nm in the spectral domain. The Harvard dataset contains 50 real-world HSIs that are captured by a commercial hyperspectral camera. Each image in the Harvard dataset is of size 1392×1040 in the spatial domain and has 31 consecutive spectral bands ranging from 420 to 720 nm at an interval of 10 nm. For the CAVE dataset, we treated the first 20 HSIs for training and the last 12 HSIs for testing; for the Harvard dataset, we used the first 30 HSIs for training and the last 20 HSIs for testing. In Fig. 3, we show an overview of the datasets used in simulation data experiments by the corresponding RGB images of HSIs.

To synthesize the LR-HSIs with different spatial degradations, we applied 8×8 and 16×16 Gaussian filters with standard deviations randomly sampled in the range $[0.5, 3.0]$ to the HR-HSIs followed by a downsampling operator with a scaling factor of 8 and 16, respectively. To simulate degradation in the spatial domain, the spectral response of a Nikon D700 camera was used to generate the HR-MSI. Similar spectral downsampling was also used in [13] and [19]. The sizes of the LR-HSI patches and HR-MSI patches are set to $\frac{96}{s} \times \frac{96}{s} \times 31$ and $96 \times 96 \times 3$ respectively for training($s = 8, 16$). Meanwhile, data augmentation techniques (e.g., flipping and rotation) are applied to generate more training data. The batch size is set to 4 and the number of iterations is set

TABLE II

AVERAGE QUANTITATIVE RESULTS OF THE PROPOSED METHODS WITH DIFFERENT NUMBER OF STAGES IN THE CAVE AND HARVARD DATASET AND SCALING FACTOR 8(DC DENOTES DENSE CONNECTION)

	CAVE						Harvard					
No. of stages	1	2	3	3(w/o DC)	4	5	1	2	3	3(w/o DC)	4	5
PSNR	42.79	43.11	43.56	43.32	43.58	43.47	42.08	42.44	42.58	42.36	42.54	42.55
SSIM	0.990	0.992	0.994	0.993	0.994	0.994	0.985	0.988	0.989	0.988	0.989	0.989
SAM	3.51	3.40	3.36	3.37	3.36	3.37	4.60	4.53	4.52	4.54	4.53	4.52
ERGAS	0.85	0.83	0.80	0.81	0.80	0.80	1.33	1.30	1.28	1.30	1.29	1.28

to 10^6. The hyperparameter α is empirically set to 30. And the number of the iteration stage and the STLs is set to 4 and 2. According to our experiment, disabling parameter sharing of the prior network across different iterations does not lead to any noticeable performance gain at the price of much higher training complexity; therefore, we propose to share parameters across stages. Meanwhile, the sharing of parameters can be interpreted as an effective regularization strategy. Adam optimizer is used to optimize network parameters, with its default setting and an initial learning rate of 1e-4. All parameters of the convolutional filters are initialized by the Xavier initializer. The proposed model is implemented on PyTorch and trained with 2 Nvidia GTX 1080Ti.

B. Comparison Methods and Evaluation Metrics

We have compared the proposed method with several representative methods, including the subspace regularized method (HySure [18]), the blind method based on the prediction of degradation (DBSR [11]), the unsupervised adaptation learning method (UAL [13]), the coupled mixing method (CuCaNet [19]). The DBIN method of [25] trained from paired LR-HSI and HR-HSI cannot be used directly in our blind HIF comparison study, as there are no HR-HSIs in our training dataset. To facilitate comparison with DBIN [25], we used the loss function of Eq. (14) to retrain this method. In particular, we applied a Softmax to the parameters of the convolution kernel to simulate the degradation process, further improving the performance of the DBIN method [25]. The peak signal-to-noise ratio (PSNR), the structural similarity index (SSIM), the spectral angle mapper (SAM), and the relative dimensionless global error in synthesis (ERGAS) are used to evaluate the objective performance of the proposed method. PSNR computes the average error difference between the reconstructed HSI and the ground truth, while SSIM calculates the spatial similarity between the reconstructed results and the ground truth.

SAM and ERGAS measure the spectral angle between pixels of the generated HSI and the ground truth and spectral quality, respectively. It is considered to gain better fusion performance when there is a smaller SAM/ERGAS and a larger PSNR/SSIM.

C. Ablation Studies

In this subsection, several ablation studies are carried out to demonstrate the effectiveness of different settings in the modules we proposed, including the number of stages, the use of dense connections in the overall architecture, the use of $\boldsymbol{\eta}_{lr}$ and $\boldsymbol{\eta}_{rgb}$, the use of the $\mathbf{W}$ estimation and LD model in the reconstruction module, and so on. First, we verify the impact of the number of iterations (that is, the number of stages T) on the performance of the proposed method. The average results on the CAVE and Harvard test dataset by the variants of the proposed model with different number of stages are shown in the Tab. II. From the Tab. II, we can see that more stages lead to a rapid improvement in PSNR performance when $T < 3$. To find a better trade-off between performance and complexity, T is set to 3 in our current implementation. Moreover, the effectiveness of the dense connection (DC) between prior networks is also shown in Tab. II, which confirmed the benefit of dense connections in stages to facilitate the flow of information. To verify the effectiveness of the backbone network and parameter sharing in the deep prior-learning module, we have implemented some comparative networks in which the Transformer-based prior module was replaced by the ResNet prior module adopted in [25] and the U-net denoising module adopted in [12] and [21]. From Tab. III, we can see that the proposed method with the prior U-net module outperformed that with the prior ResNet module by up to 0.87dB, which is consistent with the ablation study in [12]. We can also see that disabling parameter sharing of the prior module across stages does not introduce noticeable improvement on the CAVE dataset, therefore, we proposed to share parameters in prior module across iterations in our current implementation.

TABLE III

THE AVERAGE RESULTS OF PROPOSED METHODS USING THE RESNET, U-NET WITH AND WITHOUT PARAMETER SHARING(PS) AND TRANSFORMER PRIOR MODULES FOR CAVE DATASETS AND SCALING FACTOR 8

prior module	PSNR	SSIM	SAM	ERGAS
ResNet	42.69	0.992	3.81	1.14
U-net	43.56	0.994	3.36	0.80
U-Net(w/o PS)	43.58	0.994	3.36	0.80
Transformer	44.22	0.995	3.28	0.78

We have also conducted experiments on the CAVE and Harvard datasets to verify the effects of components in the reconstruction module. Similarly to [34], the proposed method unfolds the MAP estimation into a deep learning network; however, the Gaussian Scale Mixture (GSM) model is introduced into the MAP estimation in [34], and the variance of the degraded image is a learnable scalar, ignoring the sparsity constraint and the variance of the degraded images vary at each pixel. To verify the advantage of our work, we have realized DGSM in our HIF network framework, then unfolding it into a deep network, which is different from our work is

TABLE IV
ABLATION STUDY OF THE COMPONENTS OF THE RECONSTRUCTION MODULE FOR THE CAVE AND HARVARD DATASETS AND SCALING FACTOR 8

	CAVE		Harvard	
introduced prior	LD	GSM [34]	LD	GSM [34]
PSNR	43.56	42.97	42.58	41.83
SSIM	0.994	0.992	0.989	0.987
SAM	3.36	3.40	4.52	4.72
ERGAS	0.80	0.86	1.28	1.33
W estimation	✓	✗	✓	✗
PSNR	43.56	43.29	42.58	42.33
SSIM	0.994	0.994	0.989	0.988
SAM	3.36	3.38	4.52	4.59
ERGAS	0.80	0.83	1.29	1.30
$\boldsymbol{\eta}_{lr}$ and $\boldsymbol{\eta}_{rgb}$	✓	✗	✓	✗
PSNR	43.56	43.26	42.58	42.30
SSIM	0.994	0.993	0.989	0.988
SAM	3.36	3.39	4.52	4.56
ERGAS	0.80	0.82	1.29	1.31

L2 regularization rather than L1 regularization is imposed on the learned prior and spatially varying variance. From the Tab. IV, we can see that both spatially varying variance (i.e., $\boldsymbol{\eta}_{lr}$ and $\boldsymbol{\eta}_{rgb}$) and the introduced LD prior makes a great contribution to substantial performance improvement. Specifically, the proposed method with LD prior outperformed that of GSM prior by 0.59dB on the CAVE dataset, which attributes to the sparse regularization introduced by LD prior. And the proposed method with spatially varying variance $\boldsymbol{\eta}_{lr}$ and $\boldsymbol{\eta}_{rgb}$ outperformed the one without it by 0.30dB on the CAVE dataset, which illustrates the superiority of the pixel-varying variance of the input images. In addition, the effect of the introduction of the learned prior's variance (i.e. **W**) can also be shown in the tab. IV, the PSNR result with **W** estimation gains over 0.27dB and 0.25dB on the CAVE and Harvard datasets, respectively, which is credited to the faculty of the prior **W** in salient texture maintenance and degradation suppression.

To further explore what makes our method perform better compared to other competing methods in blind HIF, we conducted an experiment to discuss the contribution of the observation module. First, we have studied the difference carried by the softmax layer added to guarantee the non-negative and normalization constraints of the degradation model. Tab. VI shows that the implementation with the softmax layer outperforms that without the softmax layer by 9.22dB in the PSNR for the CAVE dataset and 6.89dB in the PSNR for the Harvard dataset. In addition, to compare the generation method of blur kernels in the spatial domain with that in previous work, we have designed the one with the blur kernel as part of network parameters, which is updated by network training instead of input-dependent blur kernels learned from the input LR-HSIs in our work. It can also be shown in the Tab. VI that the method with input-adaptive spatial degradation gained over the one with uniform spatial degradation by 0.45dB and 0.30dB in CAVE and Harvard dataset for the scaling factor is 8, respectively. This is mainly attributed to the design of the input-adaptive spatial degradation allows the model to learn the invalid spatial degradation for each input. It also can prove that the success achieved in previous blind HIF depends more on the great robustness of the proposed model, rather than on the design that coincides with reality. To further explore the relationship between the spatial blurring kernel size and performance result, the ablation study on the effects of the size of the spatial kernel has been conducted and the result can be seen in Table V. From Table V, we can see that the bigger size of spatial blurring kernel leads to an improvement in the PSNR performance when the kernel size < 14 and the PSNR results with 14×14 kernel size gain over 8×8 by 0.14dB and 0.09dB for our method with U-Net and Swin-Transformer as prior-learning module respectively, although 8×8 Gaussian filters were applied to the HR-HSIs for simulating spatial blur degradation when the scaling factor is 8. While the model with 14×14 as the spatial blurring kernel size has a comparable performance with the one with 16×16 spatial blurring kernel. Therefore, the size of the spatial blurring kernel is empirically set to 14×14 in this paper.

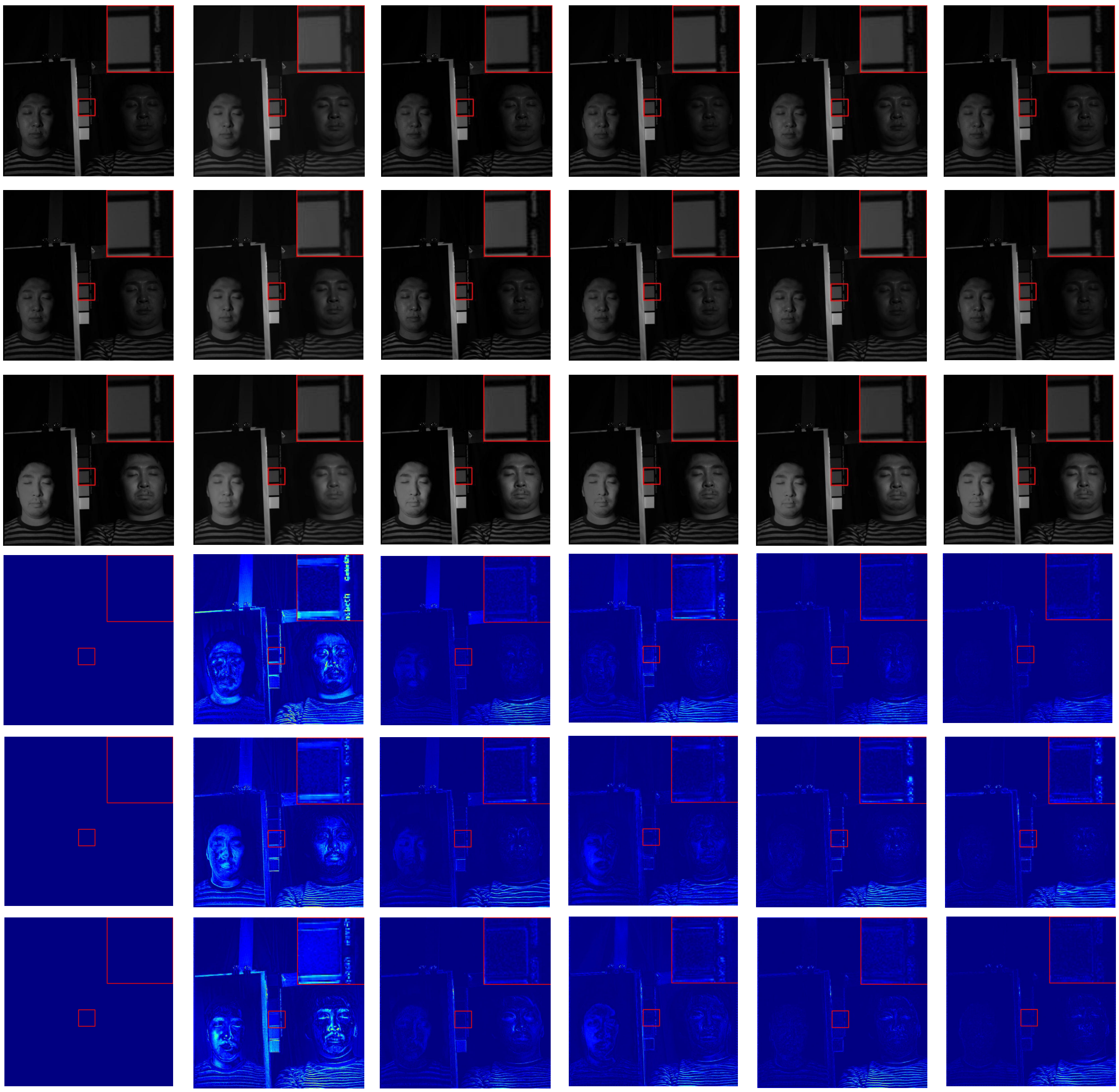
The ablation study on the learned and given ground truth spectral degradation has also been conducted and the comparison result can be seen in Tab. VI. From Tab. VI, it can be observed that the proposed network with the given spectral degradation outperforms the one with learned spectral degradation by 0.22 dB and 0.14dB on CAVE and Harvard datasets, respectively. However, it should be noticed that the given spectral degradation is completely accurate, which serves as an upper bound of the learned spectral degradation. Moreover, the training data typically consists of only LR-HSIs, HR-MSIs and HR-HSIs, without including the ground truth spectral degradation. In the case where the ground truth spectral degradation is not available, our method with the learned spectral degradation still yields comparable results as shown in Table VI, which further demonstrates the effectiveness of our method in learning the spectral degradation. Therefore, we proposed to learn the spectral degradation in the paper.

D. Performance on the CAVE and Harvard DataSets

For fairness, all deep learning-based methods have been re-trained on the same training dataset, and the result of HySure [18] and NSSR [6] were generated with the original source codes from the authors' website. For testing, we conducted experiments on the CAVE and Harvard test datasets with Gaussian blur kernels with different standard deviations outside the range [0.5, 3.0], and the average results of PSNR, SSIM, SAM, and ERGAS in the CAVE and Harvard data sets are shown in the tab. I. We have included the results of the proposed method with both U-Net and transformer modules in both datasets. From Tab. I, we can see that the proposed method performs better than all other competing methods on both CAVE and Harvard datasets when the scaling factor is 8. It should be mentioned that UAL [13] needs to be trained with paired degraded images, which introduce some information about the degradation model and NSSR [6] is nonblind, which is unfair to other competing methods. Despite all this, our method still outperforms competing methods. The average PSNR gains over the NSSR [6] method (the second best) by up to 3.02 dB on the CAVE and 1.14 dB on Harvard, as the Harvard data set is less challenging. Tab. I shows

TABLE V
ABLATION STUDY ON THE EFFECTS OF THE SIZE OF THE SPATIAL KERNEL FOR CAVE DATASET

	Ours(UNet)					Ours(Trans)				
kerne size	8*8	10*10	12*12	14*14	16*16	8*8	10*10	12*12	14*14	16*16
PSNR	43.42	43.44	43.49	43.56	43.56	44.13	44.15	44.16	44.22	44.24
SSIM	0.992	0.993	0.993	0.993	0.993	0.994	0.995	0.995	0.995	0.995
SAM	3.40	3.38	3.38	3.36	3.36	3.32	3.30	3.29	3.28	3.28
ERGAS	0.83	0.81	0.81	0.80	0.80	0.79	0.78	0.78	0.78	0.78



(a) Original (b) HySure (c) CUCaNet (d) DBSR (e) UAL (f) Ours

Fig. 4. Qualitative results of several unsupervised HIF methods performed on *photo_and_face* in the CAVE dataset at 480nm, 560nm, and 670nm with scaling factor $s = 8$. Top three rows: resulting HR-HSI imagesat 480nm, 560nm and 670nm; bottom row: heat map of error images by competing methods. (a). The ground truth image; (b).HySure [18] (PSNR = 35.66dB / SSIM = 0.944); (c).CUCaNet [19] [19] (PSNR = 40.58dB/SSIM = 0.987); (d).DBSR [11] (PSNR = 39.79dB / SSIM = 0.978); (e).UAL [13] (PSNR = 42.33dB/SSIM = 0.989); (f). Ours (PSNR = **42.93**dB/SSIM = **0.992**).

that the leading gap between our method and the second-best competing method in PSNR has been reduced to 1.14 dB for the CAVE dataset when the scaling factor is 16, even for the Harvard dataset, NSSR [6] outperforms our method by 0.15 dB in the case where the scaling factor is 16, although under unfair conditions (NSSR [6] is a nonblind method).

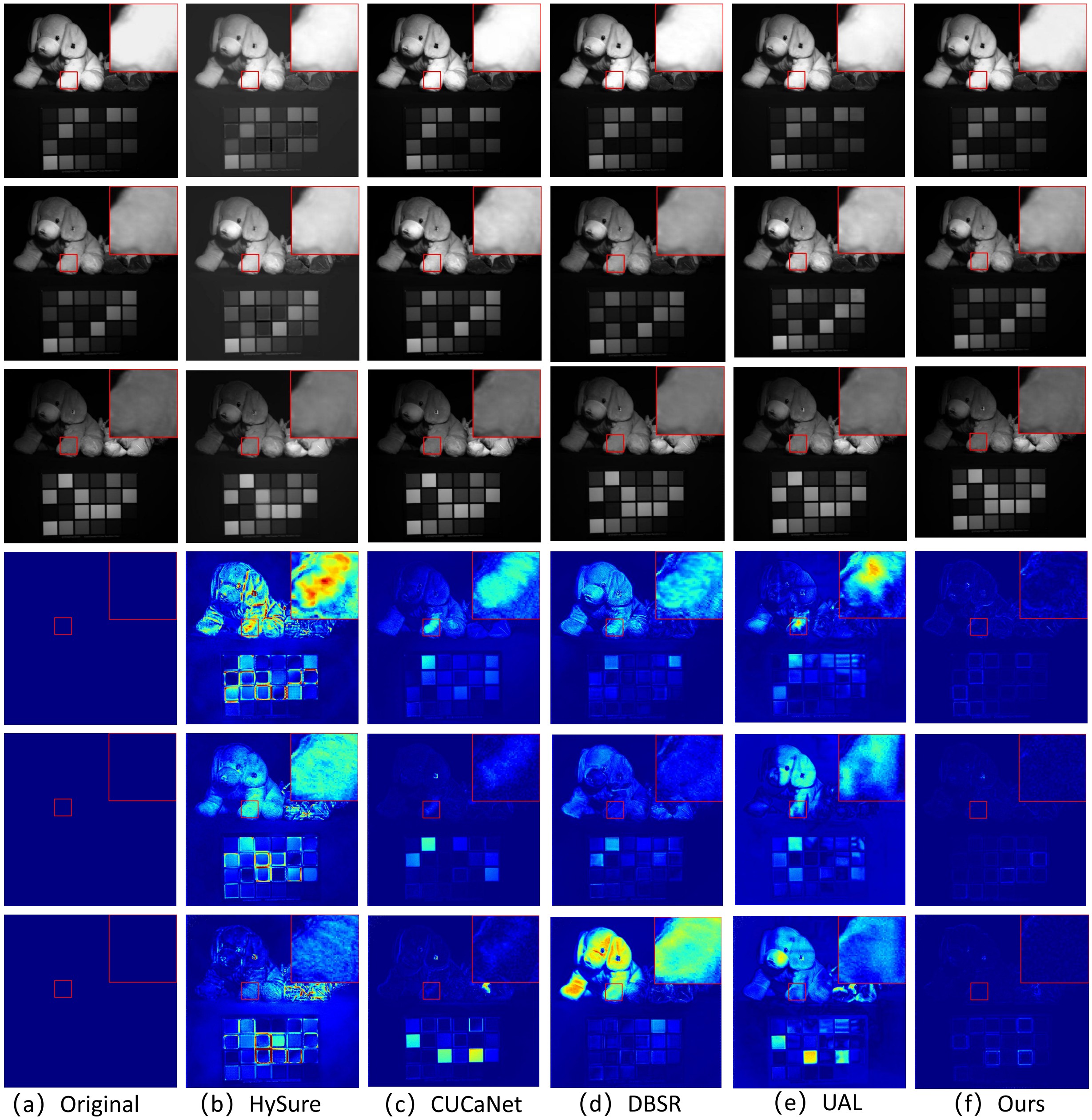


Fig. 5. Qualitative results of several unsupervised HIF methods perform on $stuffed\_toys$ in the CAVE dataset at 480nm, 560nm, and 670nm with scaling factor $s = 8$. Top three rows: resulting HR-HSI imagesat 480nm, 560nm and 670nm; bottom row: heat map of error images by competing methods. (a) Ground truth image; (b) HySure [18](PSNR = 33.56dB/SSIM = 0.914); (c) CUCaNet [19] (PSNR = 41.31dB/SSIM = 0.990); (d) DBSR [11] (PSNR = 38.93dB/SSIM = 0.979); (e) UAL [13](PSNR = 39.89dB/SSIM = 0.986); (f) Ours(PSNR = **44.38**dB/SSIM = **0.994**).

It is difficult to learn blur kernels of size 16×16 from LR-HSI of size $8 \times 8 \times 31$ with the scaling factor 16 in our experimental setting, which can be explained as constraints of objective conditions. Furthermore, to our best knowledge, our method is the only one that forces H and $H^\top$ to share the same parameters to ensure consistency between the two deep learning methods, which is faithful to mathematical expression. Therefore, the impact a poorly estimated blur kernel makes on the performance of our method is much greater than that of other learning methods.

The visual quality comparison also clearly favors the proposed HIF method. Parts of HR-HSIs reconstructed by different competing methods are shown in Figs. 4-7. From Figs. 4-7, we can see that the proposed method recovers more details than other competing methods in different bands (e.g., sharper eyeball, thin branches, palette). To facilitate visual inspection, we also included a comparison of the differences between the reconstructed image and the ground truth. As illustrated by the lower energy (i.e., better suppression of warm pixels) in the error images, the proposed method clearly outperforms others, especially around the edge and textured regions.

E. Performance on Real Multispectral Images

We have also conducted experiments on a real multispectral dataset WV2 (World View2). The dataset contains a pair of

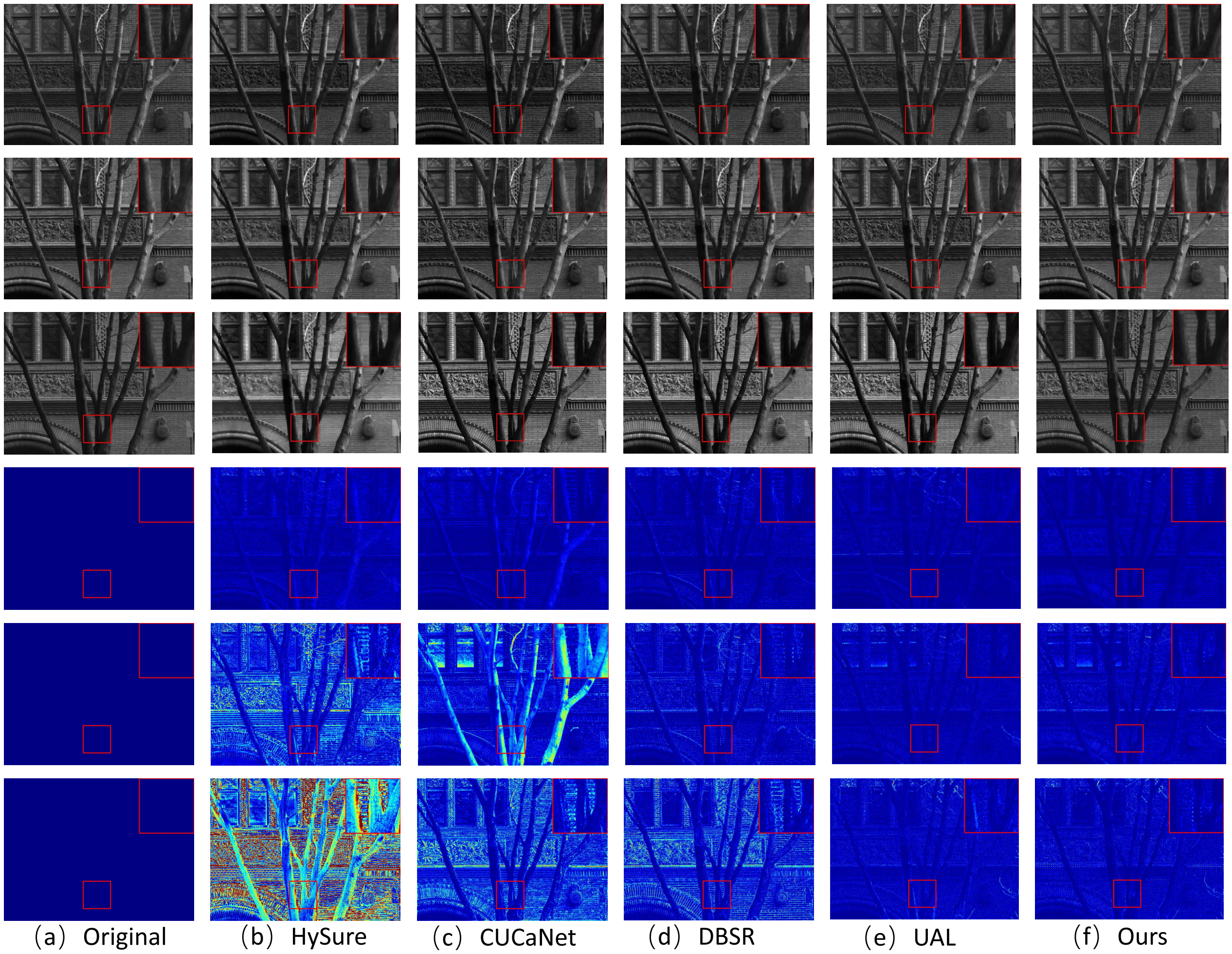


Fig. 6. Qualitative results of several unsupervised HIF methods performed on $image6$ in the Harvard dataset at 480nm, 560nm, and 670nm with scaling factor $s = 8$. Top three rows: resulting HR-HSI images at 480nm, 560nm, and 670nm; bottom row: heat map of error images by competing methods. (a) Ground truth image; (b) HySure [18] (PSNR = 34.68dB / SSIM = 0.953); (c) CUCaNet [19] (PSNR = 38.67dB/SSIM = 0.981); (d) DBSR [11] (PSNR = 37.66dB/SSIM = 0.975); (e) UAL [13] (PSNR = 41.67dB/SSIM = 0.991); (f) Ours (PSNR = **41.99**dB/SSIM = **0.994**).

TABLE VI

ABLATION STUDY OF THE COMPONENTS OF THE OBSERVATION MODULE FOR THE CAVE AND HARVARD DATASETS AND SCALING FACTOR 8

	CAVE		Harvard	
softmax layer	✓	✗	✓	✗
PSNR	43.56	34.34	42.58	35.69
SSIM	0.994	0.943	0.989	0.952
SAM	3.36	7.17	4.52	6.51
ERGAS	0.80	1.78	1.28	1.70
spatial degradation	adaptive	uniform	adaptive	uniform
PSNR	43.56	43.11	42.58	42.28
SSIM	0.994	0.992	0.989	0.986
SAM	3.36	3.43	4.52	4.61
ERGAS	0.80	0.85	1.28	1.30
spectral degradation	learned	known	learned	known
PSNR	43.56	43.78	42.58	42.72
SSIM	0.994	0.994	0.989	0.990
SAM	3.36	3.33	4.52	4.51
ERGAS	0.80	0.81	1.29	1.30

real LR-MSI of spatial size 418×658 in eight bands and an HR-RGB image of size $1676 \times 2632 \times 3$, which are used to reconstruct an HR-HSI of size $1676 \times 2632 \times 8$. Similarly to the simulated study, the upper half of LR-MSI and HR-RGB is for training, while the remaining half is for testing. Due to the design of our method, which does not need the HR-MSI label (unsupervised) and the degradation model (blind), we are not required to generate paired training data in implementing our method, while paired training data are required to pre-train parts of their networks in two recently proposed deep learning methods, DBSR [11] and UAL [13]. Since the ground truth of HR-MSI is not available, the Wald protocol [45] is used to obtain paired training samples. With Wald's protocol, the original LR-MSI and HR-RGB images are spatially degraded to generate training pairs that contain 18000 pairs of HR-RGB images of size $96 \times 96 \times 3$ and LR-MSIs of size $24 \times 24 \times 8$. We compare the proposed method with the HySure [18] method, the DBSR [11] method, and the UAL [13] method. Parts of the reconstruction results are shown in Figs. 8, from which we can see that the proposed method can recover more detail about the edges and textures of the images and fewer visual artifacts than the other method. From Figs. 8, the HySure [18] method and the DBSR [11] method fail to suppress visual artifacts more or less, while the reconstructed image of the UAL [13] method

TABLE VII
FLOPS, PARAMETERS AND PERFORMANCE OF THE COMPETING METHODS AND OUR METHOD

	TFlops	Params.(M)	PSNR	SSIM	SAM	ERGAS
DBIN	3.88	2.98	38.69	0.964	5.53	1.19
UAL	1.07	3.09	40.81	0.985	3.92	0.99
Ours(Unet-samll)	2.81	3.72	42.59	0.992	3.50	0.87
Ours(Unet)	3.66	7.37	43.46	0.994	3.36	0.80
Ours(Trans)	9.72	8.94	44.22	0.995	3.28	0.78

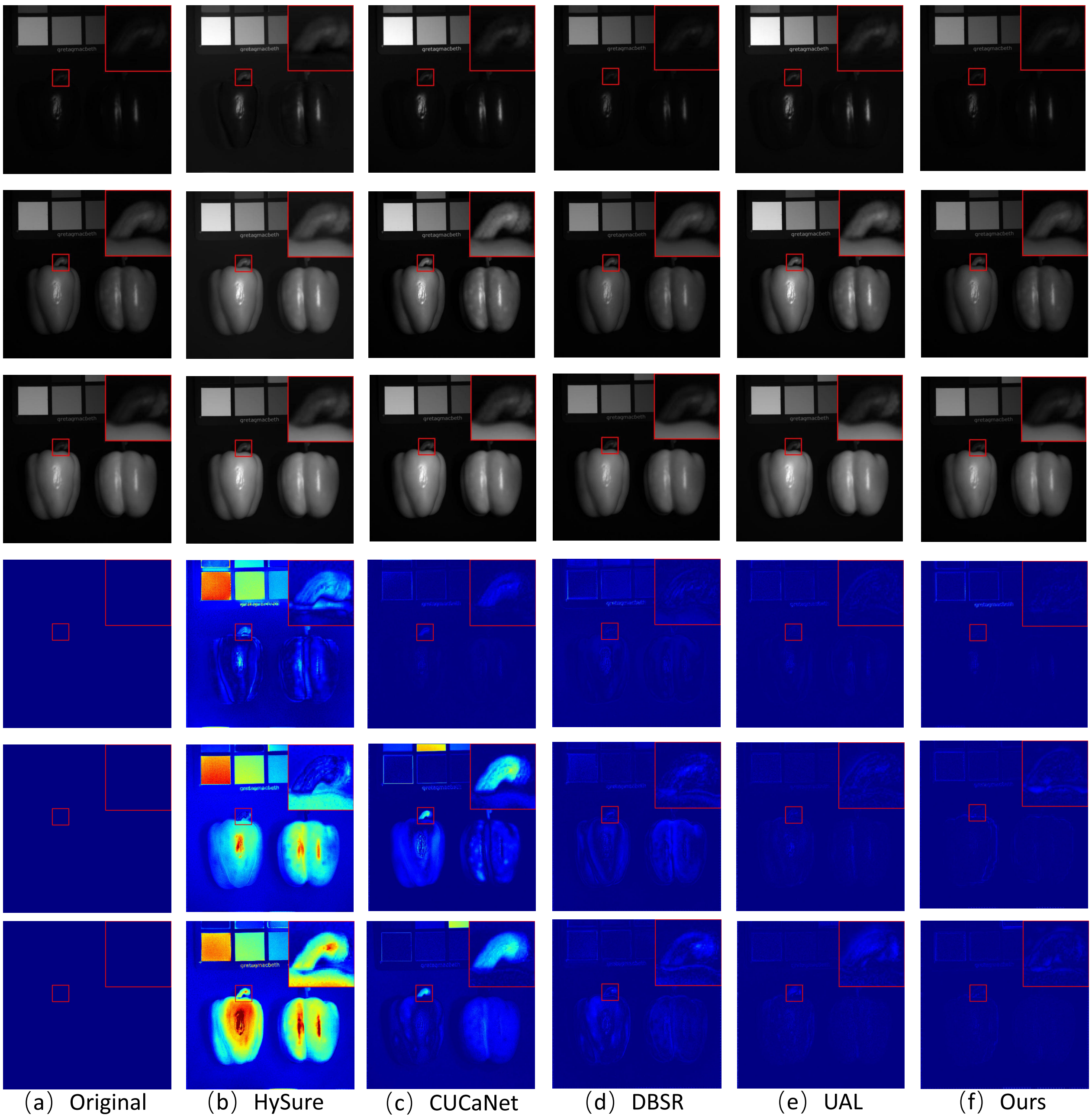


Fig. 7. Qualitative results of several unsupervised HIF methods performed on *real_and_fake_peppers* in the CAVE dataset at 480nm, 560nm, and 670nm with scaling factor $s = 8$. The top three rows: resulting HR-HSI images at 480nm, 560nm, and 670nm; bottom row: heat map of error images by competing methods. (a) Ground truth image; (b) HySure [18] (PSNR = 42.32dB / SSIM = 0.991); (c) CUCaNet [19] (PSNR = 45.38dB/SSIM = 0.994); (d) DBSR [11] (PSNR = 47.83dB / SSIM = 0.997); (e) UAL [13] (PSNR = 48.39dB/SSIM = 0.997); (f) Ours (PSNR=**49.96**dB/SSIM = **0.998**).

is obviously inferior to ours in the reconstructed details of edges and textures (e.g., the shape of the right-angled building in Fig. 8). To summarize, when the degradation model is completely unknown, the demand for paired data to pre-train

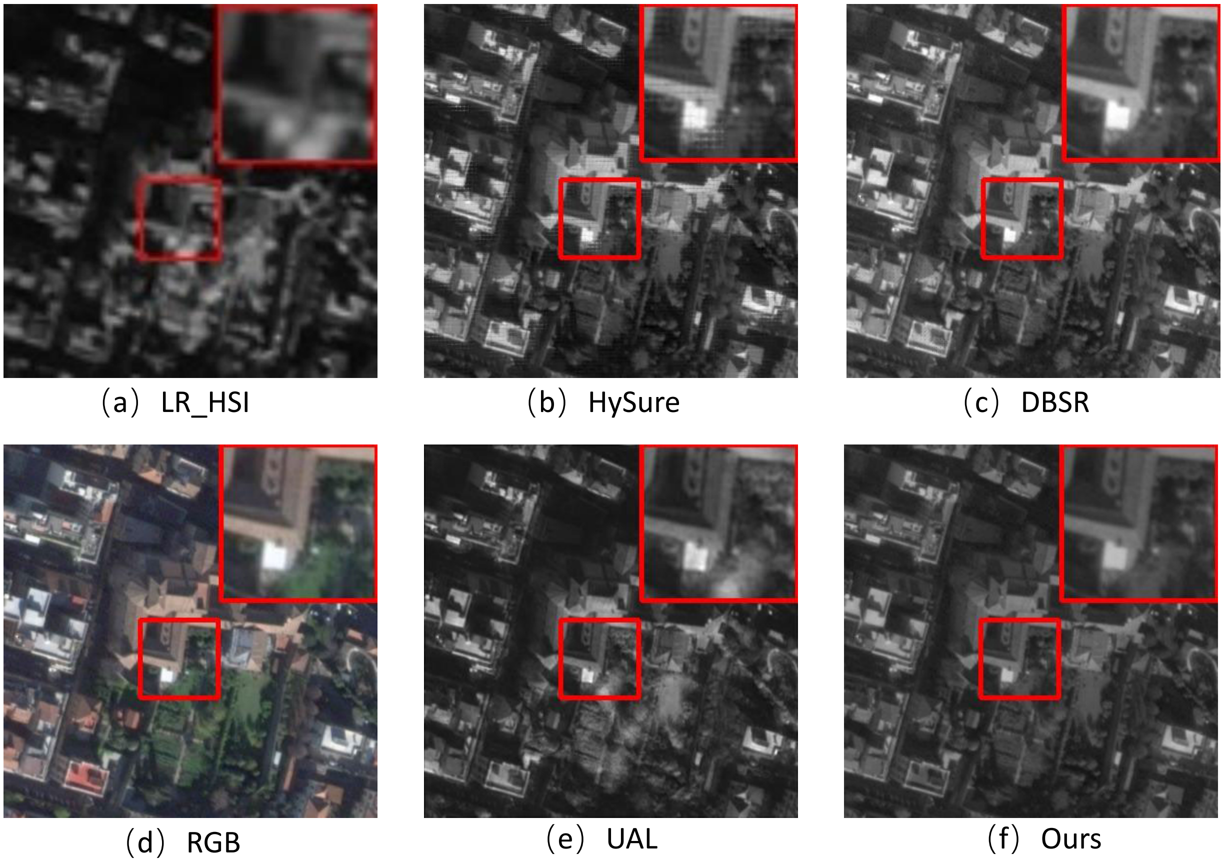


Fig. 8. Reconstruction results of several unsupervised HIF methods performed on a real multispectral image WV2 in band 5. (a) Real LR-HSI; (b) HySure [18]; (c) DBSR [11]; (d) HR-RGB image; (e) UAL [13]; (f) Ours.

the methods of UAL [13] and DBSR [11] is difficult to be satisfied, while our method utilizes the information of the physical model deeply so that the degradation model can be learned through end-to-end training. Furthermore, our method appears to perform better in suppressing visual artifacts and delicate reconstruction, judging by the results of the visual reconstruction.

F. Complexity Analysis

In addition to the comparison of quantitative and qualitative results, a comparison of complexity has also been provided in Tab. VII. From Tab. VII, though our method outperforms the UAL method [13] by up to 3.41dB, larger model params number and FLOPS seem to be unacceptable. Therefore, we have also implemented a more lightweight variant of the proposed method (denoted as Ours(Unet-small) in Tab. VII), in which the number of feature channels is set to 64 for convolutional layers in the prior-learning module. From Tab. VII, the proposed lightweight variant outperforms the UAL method by 1.68dB with comparable FLOPS and number of parameters. It can also be observed that the parameter in the prior-learning module can be significantly reduced without significant performance loss, again proving that the deep unfolding strategy plays an important role in our method.

VI. Conclusion

In this paper, we first formulate the blind HIF problem as a maximum a posteriori probability (MAP) to learn the degradation models in the spectral and spatial domains and the reconstructed HR-HSI. To address the challenge with a spatially varying blur, we introduce an observation model to learn the input-dependent blur kernels from LR-HSIs. Meanwhile, a SoftMax nonlinearity layer is applied to enforce the non-negative and normalization constraints. The derived MAP model with the learned Laplace distribution(LD) priors, by which sparse regularization is implicitly introduced into the network, then unfolded it into a multistage network implementation. Instead of a hand-crafted prior, we introduce a transformer-based network to learn both the local mean and variance priors of the LD model. Meanwhile, the introduction of the degraded images' variance on a pixel-by-pixel basis leads to an improvement in reconstruction performance. Compared to hand-crafted priors, all parameters in our method of the MAP estimator and the DCNN, including the Transformer-based prior network and the observation module, are jointly optimized through end-to-end training. Extensive experimental results on both synthetic and real datasets demonstrate that the proposed method outperforms existing state-of-the-art methods.

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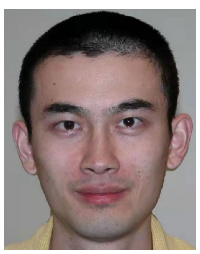

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